

AI & Machine Learning – Task 2 Report

Feature Engineering, Model Optimization & Performance Comparison

Submitted By

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Objective

The objective of this project is to improve machine learning model performance using feature engineering techniques, data preprocessing, and model comparison. The project aims to identify the most suitable regression algorithm for predicting house prices using the California Housing Dataset.

Dataset Description

The California Housing Dataset was used in this project. This dataset contains housing-related information collected from districts in California.

Target Variable

HousePrice (Median House Value)

Input Features

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Households
- Latitude
- Longitude

The dataset was loaded using the Scikit-learn library.

Data Preprocessing

Data preprocessing is an important step in machine learning. In this project, feature scaling was performed using the **StandardScaler()** technique.

Feature scaling helps in:

- Bringing all features to a common scale.
- Improving model stability.

- Reducing bias caused by features with larger values.

After preprocessing, the dataset was divided into training and testing sets.

Training Data : **80%**

Testing Data : **20%**

Random State : **42**

Machine Learning Models Used

Three regression algorithms were implemented and compared.

1. Linear Regression

Linear Regression was used as a baseline model to predict house prices.

Results

RMSE : **0.745581**

R² Score : **0.575788**

2. Ridge Regression

Ridge Regression was used to reduce overfitting by applying regularization.

Results

RMSE : **0.745554**

R² Score : **0.575819**

3. Decision Tree Regressor

Decision Tree Regressor was used to capture non-linear relationships present in the dataset.

Results

RMSE : **0.724234**

R² Score : **0.599732**

Model Performance Comparison

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819

Model	RMSE	R ² Score
Decision Tree Regressor	0.724234	0.599732

Best Model Selection

Based on the comparison results, the **Decision Tree Regressor** performed better than the other models.

Reasons:

- Lowest RMSE value (**0.724234**)
- Highest R² Score (**0.599732**)

Therefore, the Decision Tree Regressor was selected as the best-performing model for this project.

Visualization

An **Actual vs Predicted House Prices** scatter plot was generated to evaluate prediction quality.

Observations:

- Most data points are concentrated around the reference line.
 - The model captures the overall trend reasonably well.
 - Some outliers are present, which slightly affect prediction accuracy.
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Conclusion

In this project, feature scaling, preprocessing, model training, and performance comparison techniques were successfully implemented.

Among all tested algorithms, the **Decision Tree Regressor** achieved the best predictive performance with the highest R² score and lowest RMSE value.

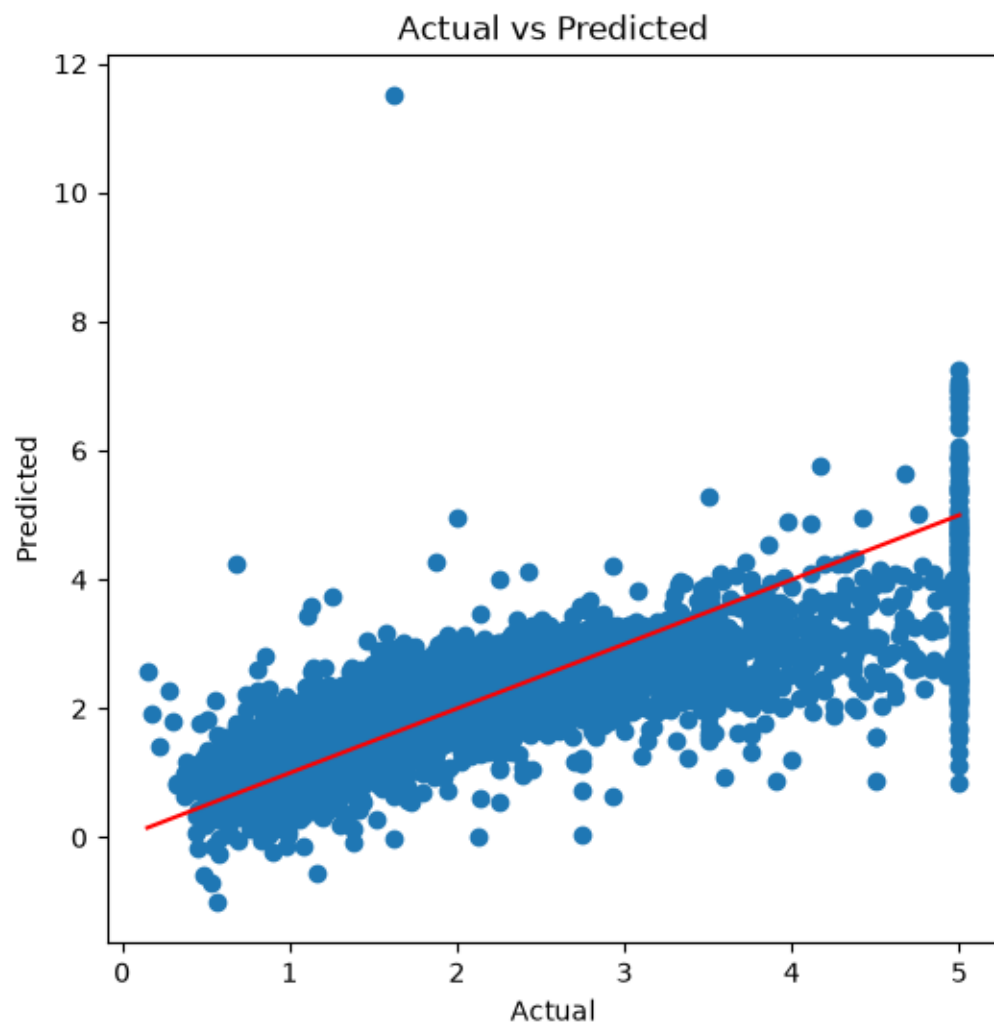
This project provided practical exposure to real-world machine learning workflows, including data preprocessing, model optimization, evaluation, and visualization techniques.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

- Jupyter Notebook
 - VS Code
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Actual vs Predicted Graph



Project Status

Task 2 Successfully Completed